

REGINA H REYNOLDS

★ KEY PUBLICATIONS

2023

Local genetic correlations exist among neurodegenerative and neuropsychiatric diseases¹

NPJ Parkinson's disease

- Reynolds, RH, Wagen, AZ, Lona-Durazo, F, Scholz, SW, Shoai, M, Hardy, J, Gagliano Taliun, SA, Ryten, M
- Role: Co-first author, lead analyst and corresponding author.
- Analysis² of local genetic correlations between neurodegenerative and neuropsychiatric disorders, with the aim of identifying genomic regions and genes that may drive pleiotropy.

2021

Cross-platform transcriptional profiling identifies common and distinct molecular pathologies in Lewy body disorders³

Acta Neuropathologica

- Feleke, R, Reynolds, RH, Smith, A, Tilley, B, Gagliano Taliun, SA, Hardy, J, Matthews, PM, Gentleman, S, Owen, D, Johnson, MR, Srivastava, P, Ryten, M
- Role: Co-first author and analyst.
- Transcriptomic analysis⁴ of cell-type-specific changes in the Lewy body diseases.

2020

Dystonia genes functionally converge in specific neurons and share neurobiology with psychiatric disorders⁵

Brain

- Mencacci, NE, Reynolds, RH, Garcia Ruiz, S, Vandrovcova, J, Forabosco, P, Sánchez-Ferrer, A, Volpato, V, UK Brain Expression Consortium, International Parkinson's Disease Genomics Consortium, Weale, ME, Bhatia, KP, Webber, C, Hardy, J, Botía, JA, Ryten, M
- Role: Co-first author and analyst.
- Integrative omics analysis of monogenic dystonias, with the aim of improving our understanding of the pathways driving this clinically heterogeneous group of movement disorders.

2019

Informing disease modelling with brain-relevant functional genomic annotations⁶

Brain

- Reynolds, RH, Hardy, J, Ryten, M, Gagliano Taliun, SA
- Role: First author.
- Review of conceptual advances in the generation of brain-relevant functional genomic annotations and among tools that allow integration of these annotations with genome-wide association summary statistics.

View this publications list online with links at <https://rhreynolds.github.io/cv>

CONTACT

✉ rhreynolds@hotmail.co.uk

🐙 [GitHub](#)

in [LinkedIn](#)

📖 [ResearchGate](#)

The long-form version of my CV is available [here](#).

The source code is available [GitHub](#).

Last updated on 2026-08-19.

- 2019 **Moving beyond neurons: the Role of cell type-specific gene regulation in Parkinson's disease heritability⁷**
NPJ Parkinson's disease
- Reynolds, RH, Botía, JA, Nalls, MA, International Parkinson's Disease Genomic Consortium (IPDGC), System Genomics of Parkinson's Disease (SGPD), Hardy, J, Gagliano Taliun, SA, Ryten, M
 - Role: First author and lead analyst.
 - Analysis of Parkinson's disease common variation, with the aim of identifying cell types and pathways of importance to disease risk.
- 2018 **Perturbations in the p53/miR-34a/SIRT1 pathway in the R6/2 Huntington's disease model⁸**
Molecular and Cellular Neuroscience
- Reynolds, RH, Petersen, MH, Willert, CW, Heinrich, M, Nymann, N, Dall, M, Treebak, JT, Björkqvist, M, Silahatoglu, A, Hasholt, L and Nørremølle, A
 - Role: First author, lead experimentalist and analyst.



PUBLICATIONS

- 2025 **Splicing accuracy varies across human introns, tissues, age and disease⁹**
Nature Communications
- Garcia-Ruiz, S, Zhang, D, Gustavsson, EK, Rocamora-Perez, G, Grant-Peters, M, Fairbrother-Browne, A, Reynolds, RH et al.
 - Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
- 2025 **Astrocytic RNA editing regulates the host immune response to alpha-synuclein¹⁰**
Science Advances
- D'Sa, K, Choi, ML, Wagen, AZ, Seto-Savlia, N, Kopach, O, Evans, JR, Rodrigues, M, Lopez-Garcia, P, Lachica, J, Clarke, BE, Singh, J, Ghareeb, A, Bayne, J, Grant-Peters, M, Garcia-Ruiz, S, Chen, Z, Rodrigues, S, Athauda, A, Gustavsson, EK, Gagliano-Taliun, SA, Toomey, C, Reynolds RH et al.
 - Role: Analyst and adviser. Contributed to tooling around alignment, advised on deconvolution and RNA-editing. Critiqued manuscript.
- 2024 **The annotation of GBA1 has been concealed by its protein-coding pseudogene GBAP1¹¹**
Science Advances
- Gustavsson, EK, Sethi, S, Gao, Y, Brenton, J, Ruiz, SG, Zhang, D, Garza, R, Reynolds, RH et al.
 - Role: Adviser. Critiqued manuscript.
- 2024 **Genome sequence analyses identify novel risk loci for multiple system atrophy¹²**
Neuron
- Chia, R, Ray, A, Shah, Z, Dong, J, Ruffo, P, Fujita, M, Menon, V, Saez-Atienzar, S, Reho, P, Kaivola, K, Walton, R, Reynolds RH et al.
 - Role: Adviser. Reviewed and critiqued colocalisation analyses.

2023	The non-specific lethal complex regulates genes and pathways genetically linked to Parkinson's disease¹³ Brain • Hicks, AR, Reynolds, RH et al. • Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2023	ensemblQueryR: fast, flexible and high-throughput querying of Ensembl LD API endpoints in R¹⁴ GigaByte • Fairbrother-Browne, A, Garcia-Ruiz, S, Reynolds, RH et al. • Role: Adviser. Contributed to conceptualisation and critiqued manuscript.
2023	Analysis of subcellular RNA fractions demonstrates significant genetic regulation of gene expression in human brain post-transcriptionally¹⁵ Scientific Reports • D'Sa, K, Guelfi, S, Vandrovcova, J, Reynolds RH et al. • Role: Adviser. Contributed to troubleshooting of analyses.
2023	aws-s3-integrity-check: an open-source bash tool to verify the integrity of a dataset stored on Amazon S3¹⁶ GigaByte • Garcia-Ruiz, S, Reynolds, RH et al. • Role: Adviser. Contributed to idea development and critiqued manuscript.
2023	Genome-wide Analysis of Motor Progression in Parkinson Disease¹⁷ Neurology Genetics • Martinez-Carrasco, A, Real, R, Lawton, M, Reynolds, RH et al. • Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2023	Regional genetic correlations highlight relationships between neurodegenerative disease loci and the immune system¹⁸ Communications Biology • Lona-Durazo, F, Reynolds, RH et al. • Role: Analyst and advisor. Contributed to code base and critiqued manuscript.
2023	Association between the LRP1B and APOE loci and the development of Parkinson's disease dementia¹⁹ Brain • Real, R, Martinez-Carrasco, A, Reynolds, RH et al. • Role: Adviser. Critiqued manuscript.
2023	The contribution of Neanderthal introgression and natural selection to neurodegenerative diseases²⁰ Neurobiology of Disease • Chen, Z, Reynolds RH et al. • Role: Author and analyst. Performed partitioned SNP-heritability analyses and contributed to writing of results.

2023	The IPDGC/GP2 Hackathon – an open science event for training in data science, genomics, and collaboration using Parkinson’s disease data²¹ NPJ Parkinson’s disease · Leonard, HL, Reynolds, RH et al. on behalf of The International Parkinson Disease Genomics Consortium (IPDGC) and The Global Parkinson’s Genetics Program (GP2) · Role: Author and analyst. Contributed to creation of tools.
2022	IntroVerse: a comprehensive database of introns across human tissues²² Nucleic Acid Research · García-Ruíz, S, Gustavsson, EK, Zhang, D, Reynolds, RH et al. · Role: Adviser. Critiqued manuscript.
2022	Genome-wide association study of REM sleep behavior disorder identifies polygenic risk and brain expression effects²³ Nature Communications · Krohn, L, Heilbron, K, Blauwendraat, C, Reynolds, RH, et al. · Role: Analyst. Performed colocalisation analyses and gene expression analyses.
2022	ggtranscript: an R package for the visualization and interpretation of transcript isoforms using ggplot2²⁴ Bioinformatics · Gustavsson, EK, Zhang, D, Reynolds, RH, Ruiz, SG, Ryten, MR · Role: Adviser. Performed beta testing and critiqued manuscript.
2021	A genetic link between risk for Alzheimer’s disease and severe COVID-19 outcomes via the OAS1 gene²⁵ Brain · Magusali, N, Graham, AC, Piers, TM, Panichnantakul, P, Yaman, U, Shoi, M, Reynolds, RH et al. · Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2021	Mitochondrial-nuclear cross-talk in the human brain is modulated by cell type and perturbed in neurodegenerative disease²⁶ Communications Biology · Fairbrother-Browne, A, Ali, A, Reynolds, RH et al. · Role: Adviser. Advised on tools available for cell-type specificity and heritability analyses.
2021	A systems-level analysis highlights microglial activation as a modifying factor in common epilepsies²⁷ Neuropathology and Applied Neurobiology · Altmann, A, Ryten, M, Di Nunzio, M, Ravizza, T, Tolomeo, D, Reynolds, RH et al. · Role: Analyst. Collated gene lists designed to capture microglial states.

2021	An integrated genomic approach to dissect the genetic landscape regulating the cell-to-cell transfer of a-synuclein²⁸ Cell Reports · Kara, E, Crimi, A, Wiedmer, A, Emmenegger, M, Manzoni, C, Bandres-Ciga, S, D'Sa, K, Reynolds, RH et al. · Role: Analyst. Performed expression-weighted cell-type enrichment analyses.
2021	Heritability enrichment implicates microglia in Parkinson's disease pathogenesis²⁹ Annals of Neurology · Andersen, MS, Bandres-Ciga, S, Reynolds, RH et al. · Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2021	Human-lineage-specific genomic elements: relevance to neurodegenerative disease and APOE transcript usage³⁰ Nature Communications · Chen, Z, Zhang, D, Reynolds, RH et al. · Role: Analyst. Performed partitioned SNP-heritability analyses.
2021	Identification of Candidate Parkinson Disease Genes by Integrating Genome-Wide Association Study, Expression, and Epigenetic Data Sets³¹ JAMA Neurology · Kia, DA, Zhang, D, Guelfi, S, Manzoni, C, Hubbard, L, Reynolds, RH et al. · Role: Analyst. Involved in cell-type annotation of prioritised genes.
2021	Modelling multifunctionality of genes with secondary gene co-expression networks in human brain provides novel disease insights³² Bioinformatics · Sánchez, JA, Gil-Martínez, AL, Cisterna, A, García-Ruíz, S, Gómez-Pascual, A, Reynolds, RH et al. · Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2021	Genome sequencing analysis identifies new loci associated with Lewy body dementia and provides insights into the complex genetic architecture³³ Nature Genetics · Chia, R, Sabir, MS, Bandres-Ciga, S, Saez-Atienzar, S, Reynolds, RH et al. · Role: Analyst. Performed colocalisation analyses and gene expression analyses.
2021	CoExp Web, a web tool for the exploitation of co-expression networks³⁴ Frontiers in Genetics · García-Ruíz, S, Gil-Martínez, AL, Cisterna, A, Jurado, F, Reynolds, RH et al. · Role: Analyst. Involved in cell-type annotation of gene co-expression networks.

2021	Genetic determinants of survival in progressive supranuclear palsy: a genome-wide association study³⁵ Lancet Neurology • Jabbari, E, Koga, S, Valentino, R, Reynolds, RH et al. • Role: Analyst. Performed colocalisation analyses.
2021	Genome-wide association studies of cognitive and motor progression in Parkinson's disease³⁶ Movement Disorders • Tan, MMX, Lawton, MA, Jabbari, E, Reynolds, RH et al. • Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2021	Genetic analysis of amyotrophic lateral sclerosis identifies contributing pathways and cell types³⁷ Science Advances • Saez-Atienzar, S, Bandres-Ciga, S, Langston, RG, Kim, JJ, Choi, SW, Reynolds, RH et al. • Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2020	Differences in network controllability and regional gene expression underlie hallucinations in Parkinson's disease³⁸ Brain • Zarkali, A, McColgan, P, Ryten, M, Reynolds, RH et al. • Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2020	Dementia risk in Parkinson's disease is associated with interhemispheric connectivity loss and determined by regional gene expression³⁹ NeuroImage: Clinical • Zarkali, A, McColgan, P, Ryten, M, Reynolds, RH et al. • Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2020	Analysis of brain atrophy and local gene expression in genetic frontotemporal dementia⁴⁰ Brain Communications • Altmann, A, Cash, DM, Bocchetta, M, Heller, C, Reynolds, RH et al. • Role: Analyst. Performed expression-weighted cell-type enrichment analyses.
2020	Incomplete annotation of disease-associated genes is limiting our understanding of Mendelian and complex neurogenetic disorders⁴¹ Science Advances • Zhang, D, Guelfi, S, Ruiz, SG, Costa, B, Reynolds, RH et al. • Role: Analyst. Performed expression-weighted cell-type enrichment analyses.

- 2020 **Regulatory sites for known and novel splicing in human basal ganglia are enriched for disease-relevant information⁴²**
Nature Communications
- Guelfi, S*, D'Sa, K*, Botía, JA*, Vandrovцова, J, Reynolds, RH et al.
 - Role: Experimentalist and analyst. Performed PCR validation of transcription and partitioned SNP-heritability analyses.
- 2019 **Genetic variability in response to amyloid beta deposition influences Alzheimer's risk⁴³**
Brain Communications
- Salih, DA, Bayram, S, Guelfi, S, Reynolds, RH et al.
 - Role: Analyst. Performed colocalisation analyses.
- 2019 **Mitochondria function associated genes contribute to Parkinson's Disease risk and later age at onset⁴⁴**
NPJ Parkinson's disease
- Billingsley, KJ, Barbosa, IA, Bandrés-Ciga S, Quinn, JP, Bubb, VJ, Deshpande, C, Botía, JA, Reynolds, RH et al.
 - Role: Analyst. Performed expression-weighted cell-type enrichment analyses.
- 2018 **Variation at the TRIM11 locus modifies Progressive Supranuclear Palsy phenotype⁴⁵**
Annals of Neurology
- Jabbari, E, John, W, Tan, MMX, Maryam, S, Pittman, A, Ferrari, R, Mok, KY, Zhang, D, Reynolds, RH et al.
 - Role: Analyst. Performed gene-based association testing and analysis of gene expression data.



PRE-PRINTS

- 2025 **TP53 and RB1 are predictive genetic biomarkers for sensitivity to cytarabine in gliomas⁴⁶**
bioRxiv
- Laaniste, L, Luginbuehl, J, Trinh, M, Watkins, R, Lueg, G, Neilson, G, Burgers, M, Kania, K, Reynolds, RH et al.
 - Role: Analyst. Processed and analysed pooled CRISPRi data & bulk RNA-sequencing from primary patient-derived glioma tumourspheres.
- 2025 **Detecting cell-level transcriptomic changes of Perturb-seq using Contrastive Fine-tuning of Single-Cell Foundation Models⁴⁷**
bioRxiv
- Zhao, W, Solaguren-Beascoa, A, Neilson, G, Reynolds, RH et al.
 - Role: Adviser. Advised on validation strategy and critiqued manuscript.
- 2025 **Molecular and cellular signatures differentiate Parkinson's disease from Parkinson's disease with dementia⁴⁸**
bioRxiv
- Fairbrother-Browne, A, Grant-Peters, M, Brenton, JW, Nelvagal, H, Reynolds, RH et al.
 - Role: Experimentalist and analyst. Involved in nuclei isolation and 10X library preparation. Aligned and QC-ed single-nucleus RNA-sequencing. Critiqued manuscript.

2025	17q21.31 locus regulates Parkinson's disease relevant pathways through KANSL1 activity⁴⁹ bioRxiv · Hicks, AR, O'Callaghan, B, Brenton, JW, Grant-Peters, M, Fairbrother-Browne, A, Perez, GR, Loh, CA, Reynolds, RH et al. · Role: Experimentalist and analyst. Involved in nuclei isolation and 10X library preparation. Aligned and QC-ed single-nucleus RNA-sequencing.
2024	Ancestry-specific gene expression in peripheral monocytes mediates risk of neurodegenerative disease⁵⁰ bioRxiv · Wagen, AZ, Reynolds RH et al. · Role: Adviser. Reviewed and critiqued statistical analyses and manuscript.
2021	Detection of pathogenic splicing events from RNA-sequencing data using dasper⁵¹ bioRxiv · Zhang, D, Reynolds, RH, et al. · Role: Adviser. Helped guide and troubleshoot feature selection for outlier detection of splicing events.
2019	ERASE: Extended Randomization for assessment of annotation enrichment in ASE datasets⁵² bioRxiv · D'Sa, K, Reynolds RH et al. · Role: Analyst. Performed partitioned SNP-heritability analyses.
2018	G2P: Using machine learning to understand and predict genes causing rare neurological disorders⁵³ bioRxiv · Botía, JA, Guelfi, S, Zhang, D, D'Sa, K, Reynolds, RH et al. · Role: Adviser. Helped interpret the output of machine-learning-based models to distinguish disease genes from non-disease genes.